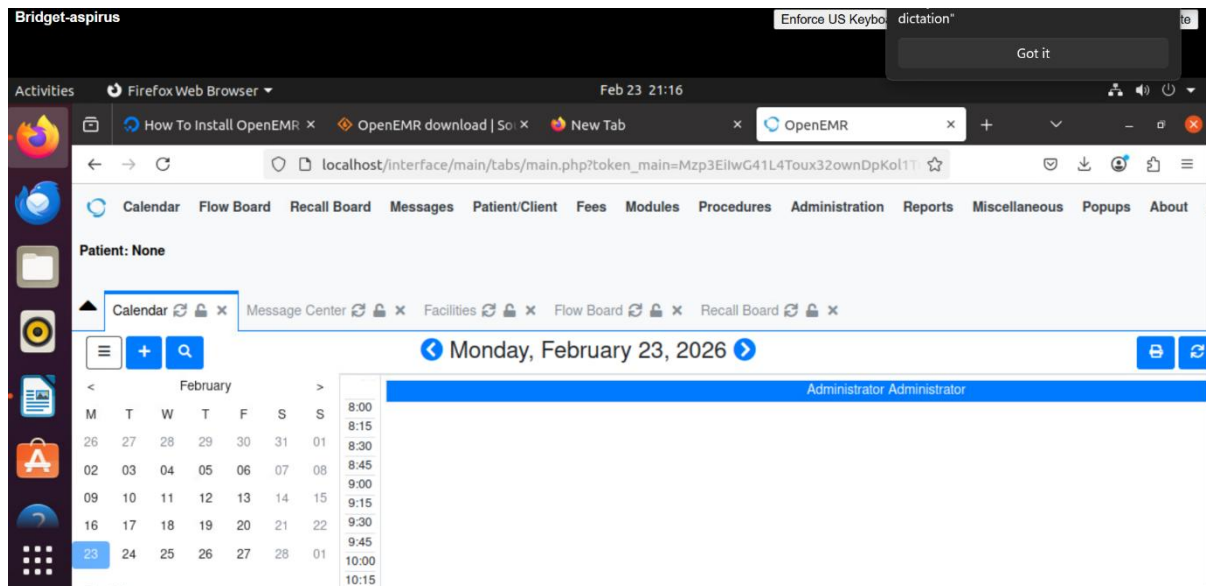


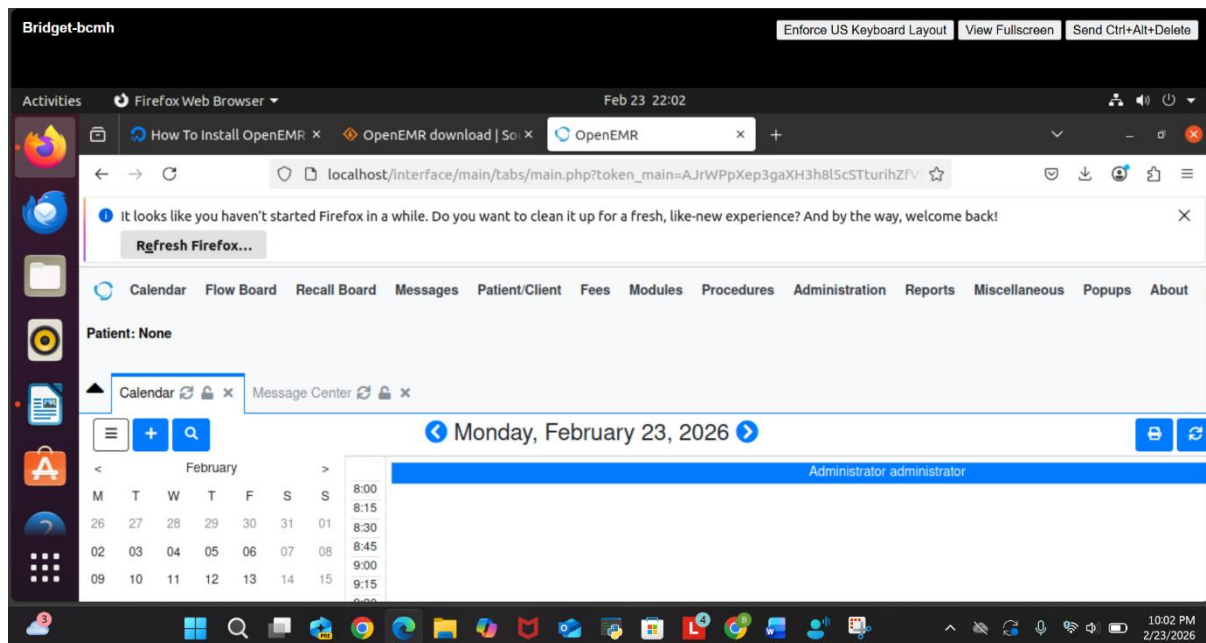
## Architecture Development Part 2: Installation, Configuration, and Security of OpenEMR

1.0. Web page screenshot (the screenshot must include the IP Address of each hospital) of each hospital's successful installation of OpenEMR (8 points)

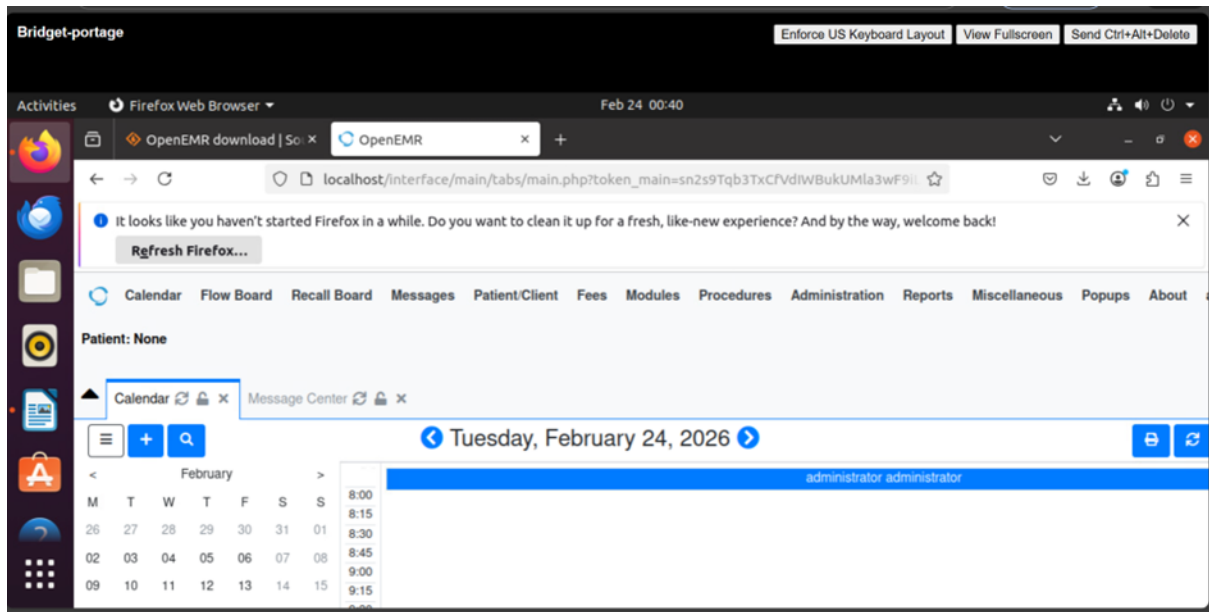
### Aspirus Hospital Webpage Screenshot(2 points)



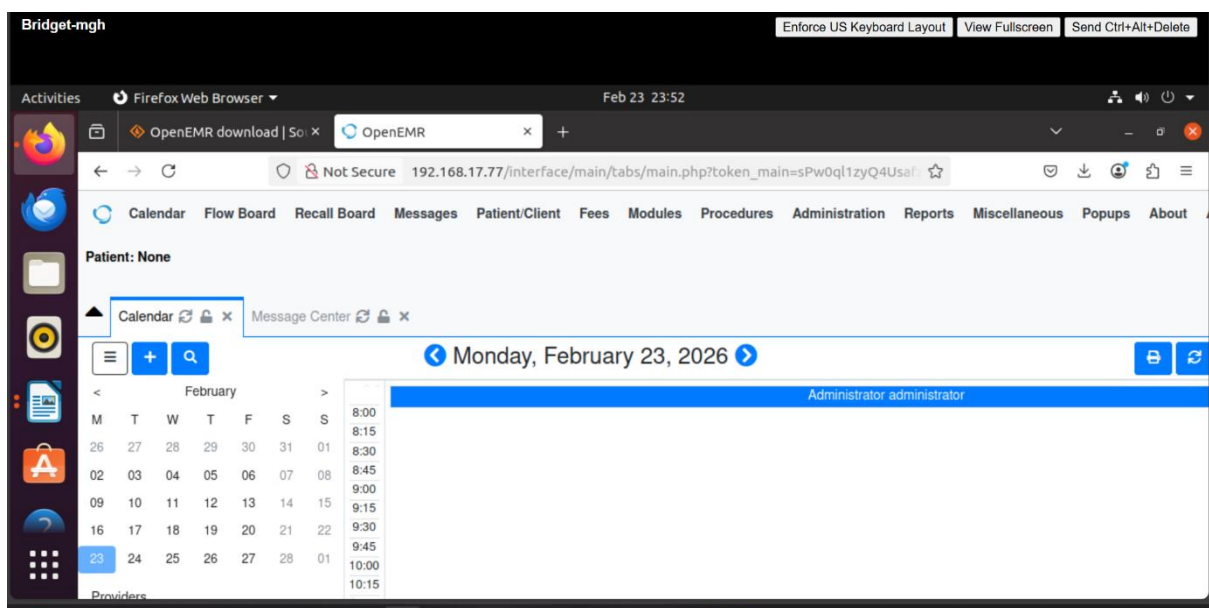
### Baraga County Memorial Hospital (BCMh) Webscreenshot (2 points)



## Portage Health Hospital Webscreenshot(2 points)



## Marquette General Hospital (MGH) Webscreenshot(2 points)



**B. Show the steps and commands you used to secure OpenEMR (10 points)**

During the installation of OpenEMR, I first ensured that my system was fully updated with the latest security patches. I ran `sudo apt-get update` followed by `sudo apt-get upgrade` to make sure all existing packages and system libraries were current and secure before installing the application.

Next, I configured automatic security updates to keep the system protected without requiring manual intervention. I installed the unattended upgrades package using `sudo apt-get install unattended-upgrades` and enabled it with `sudo dpkg-reconfigure --priority=low unattended-upgrades`. This ensured that future security patches would be applied automatically.

I then installed and configured the Uncomplicated Firewall (UFW) to control incoming and outgoing traffic. I used `sudo apt-get install ufw` to install it, allowed necessary services with `sudo ufw allow http` and `sudo ufw allow ssh`, and activated the firewall using `sudo ufw enable`. This restricted access to only essential services and reduced exposure to unauthorized connections.

To further secure the web server, I hardened the Apache configuration by editing the security settings with `sudo nano /etc/apache2/conf-available/security.conf`. I enabled important security headers to protect against common web-based attacks and reduce server information exposure.

Finally, I secured the MySQL database by running `sudo mysql_secure_installation`. During this process, I set a strong root password, disabled remote root login, and removed anonymous users to prevent unauthorized database access and strengthen overall system security.

**C. What other types of attacks would still be susceptible to the OpenEMR platform? (2 points)**

OpenEMR may still be susceptible to Distributed Denial-of-Service (DDoS) attacks, where attackers flood the server with traffic to disrupt services, and brute force attacks, where repeated login attempts are used to try to gain unauthorized access.